Element	Description	Project Details (University Transport Route Tracking)
Project Sponsor	The personentity initiating the project.	Dean of Student Affairs / University Transportation Manager.
Business Needs	The business reasons for initiating the system.	• Eliminate uncertainty regarding university bus arrival times. • Reduce long waiting times at bus stops. • Improve student safety and overall daily commute satisfaction.
Business Requirements	The business capabilities that the system will provide.	• Real-time GPS tracking for all university bus routes. • Live ETA (Estimated Time of Arrival) notifications for students. • Route and schedule management dashboard for transportation admins. • Driver app to broadcast location and report route delays or breakdowns.
Business Value	The benefits that the system will create for the organization.	• Tangible: 15% reduction in fuel/fleet operational costs due to optimized routing. • Intangible: 40% increase in student satisfaction and reduced missed morning lectures.
Special Issues	Constraints on system / special issues.	Must be fully operational before the start of the upcoming academic semester; requires stable mobile network connectivity on all buses.

Part 2: Feasibility Study

Feasibility analysis guides the organization in determining whether to proceed with the project across three dimensions: Technical, Economic, and Organizational.

1. Technical Feasibility ("Can We Build It?")

* Familiarity with Application: Medium-High. Tracking systems are widely used (e.g., Uber, Google Maps APIs).

* Familiarity with Technology: High. The development team is familiar with Flutter/React Native for mobile apps, Node.js for backend, and Google Maps API for geolocation.

* Project Size: Moderate. Includes 3 main interfaces (Student App, Driver App, Admin Web Dashboard).

* Compatibility: Easily compatible with existing university infrastructure and student portal databases.

2. Economic Feasibility ("Should We Build It?")

* Development Costs (One-Time):

  * System Analysis & Software Design: $1,500

  * Application Development & Testing: $3,000

  * GPS Hardware Installation on Buses: $1,000

  * Total Development Cost: $5,500

* Annual Operating Costs:

  * Cloud Server & Database Hosting: $600/year

  * Maps API Licensing & Data Plans: $400/year

  * System Maintenance & Updates: $500/year

  * Total Annual Operating Cost: $1,500/year

* Annual Benefits:

  * Fleet Fuel & Route Efficiency Savings: $2,500/year

  * Reduction in Administrative Overhead: $1,200/year

  * Total Annual Benefit: $3,700/year

3. Organizational / Operational Feasibility ("If We Build It, Will They Use It?")

* Senior Management: Highly supportive as it enhances university services and modernization.

* Primary Users (Students): High adoption expected since students urgently need reliable transport schedules.

* Drivers: Requires minimal training (simple "Start Route / End Route" button interface on the driver app).